Ideal-PIP Relaxed-Column Discussion

1 Goal

We want a principled hierarchical model for fine-mapping with out-of-sample LD that has both of the following properties.

1. When $L = 1$, the single-effect PIP is the ideal SER PIP. In particular, because the in-sample LD matrix has diagonal entries equal to one, the PIP for SNP j should depend on the active summary statistic x_j, not on the unknown off-diagonal entries of the in-sample LD matrix.

2. When $L > 1$, the off-diagonal LD values should still matter through the iterative residual updates. We want relaxed LD columns to help remove fitted effects when the reference LD $\bar{R}$ differs from the in-sample LD.

The current SuSiE-relax idea has the second property, but its single-effect PIP is not exactly the ideal PIP. The prime variant makes the single-effect PIP closer to the desired form, but empirically the PIP can become too diffuse. The target here is to separate the two roles of LD uncertainty:

SNP selection evidence versus effect-shape estimation for residual removal.

2 Shared-R Model

A natural principled starting point is the original shared-R model, with R treated as the unknown in-sample correlation matrix:

$$R \sim p(R \mid \bar{R}), \qquad \operatorname{diag}(R) = 1,$$

$$\gamma_\ell \sim \operatorname{Unif}\{1, \dots, J\}, \qquad \beta_\ell \sim \mathcal{N}(0, \sigma_\ell^2), \qquad \ell = 1, \dots, L,$$

and

$$b = \sum_{\ell=1}^{L} \beta_\ell e_{\gamma_\ell}, \qquad x \mid R, b \sim \mathcal{N}(Rb, R/N).$$

This model is attractive because R appears in both the mean and the noise covariance. The same unknown LD matrix that shapes the signal also shapes the sampling noise.

3 Why the SER PIP Is Ideal

For $L = 1$, conditional on fixed R, compare an active model $\gamma = j$ to the null model. The likelihood ratio contribution involving β is

$$\exp\left\{N\beta R_{\cdot j}^\top R^{-1}x - \frac{N}{2}\beta^2 R_{\cdot j}^\top R^{-1}R_{\cdot j}\right\}.$$

Since

$$R_{\cdot j}^\top R^{-1}x = x_j, \qquad R_{\cdot j}^\top R^{-1}R_{\cdot j} = R_{jj} = 1,$$

the likelihood ratio simplifies to

$$\exp\left\{N\beta x_j - \frac{N}{2}\beta^2\right\}.$$

Thus the active-SNP evidence depends on R only through the diagonal entry R_{jj}, which is one. Integrating over $\beta \sim \mathcal{N}(0, \sigma_0^2)$ gives the usual SER approximate Bayes factor:

$$\mathrm{ABF}_j = (1 + N\sigma_0^2)^{-1/2} \exp\left\{\frac{N}{2}\frac{N\sigma_0^2}{1 + N\sigma_0^2}x_j^2\right\}.$$

Therefore, after also integrating over the unknown R,

$$\Pr(\gamma = j \mid x) \propto \mathrm{ABF}_j\,.$$

All off-diagonal LD uncertainty is common nuisance evidence for the single-effect PIP and cancels in the normalization over j.

4 A useful decomposition

For any correlation matrix R, define

$$S_j = R_{-j,-j} - R_{-j,j}R_{j,-j}.$$

Under the single-effect likelihood

$$x \mid R, \beta, \gamma = j \sim \mathcal{N}(\beta R_{\cdot j}, R/N),$$

we have

$$x_j \mid R, \beta, \gamma = j \sim \mathcal{N}(\beta, 1/N),$$

because $R_{jj} = 1$. Also, by the Gaussian conditioning formula,

$$\begin{aligned} x_{-j} \mid x_j, R, \beta, \gamma = j &\sim \mathcal{N}\left(\beta R_{-j,j} + R_{-j,j}(x_j - \beta), \frac{1}{N}S_j\right) \\ &= \mathcal{N}\left(x_j R_{-j,j}, \frac{1}{N}S_j\right). \end{aligned}$$

Therefore

$$p(x \mid R, \beta, \gamma = j) = \mathcal{N}(x_j \mid \beta, 1/N)\,\mathcal{N}\left(x_{-j} \mid x_j R_{-j,j}, \frac{1}{N}S_j\right). \tag{1}$$

This is the key decomposition: conditional on x_j, the distribution of the nonactive coordinates does not involve β.

5 Choice of distribution of correlation matrices

Can we have a distribution of correlation matrices that is concentrated around $\bar{R}$, with a parameter like ν controlling the concentration? The ideal-PIP argument does not require a specific distribution for R. It only requires

$$R \in \mathcal{C}_J = \{R : R = R^\top,\ R \succ 0,\ \mathrm{diag}(R) = 1\},$$

and a prior $p(R \mid \bar{R})$ that does not depend on γ or β. Therefore the main modeling problem is choosing a useful prior on $\mathcal{C}_J$.

The prior should satisfy the following desiderata.

1. It should be centered near the reference LD matrix $\bar{R}$.
2. It should have a scalar or low-dimensional concentration parameter, such as ν or ν_0, controlling how much the in-sample LD can deviate from $\bar{R}$.
3. It should allow tractable, or at least locally tractable, posterior updates for active LD columns.

One thing we cannot have exactly is a nondegenerate distribution on $\mathcal{C}_J$ with Gaussian column marginals

$$R_{-j,j} \sim \mathcal{N}(\bar{R}_{-j,j}, \bar{S}_j/\nu)$$

for all j. A Gaussian vector has support on all of $\mathbb{R}^{J-1}$, whereas a correlation column is bounded and constrained by positive definiteness. Thus Gaussian relaxed columns should be viewed as local approximations, not as exact prior marginals of a correlation-matrix distribution.

Several candidate priors are worth considering.

Standardized inverse-Wishart. Draw a covariance matrix and standardize it:

$$\tilde{R} \sim \mathcal{IW}(\nu_0 R_0, \nu_0 + J + 1), \qquad R = \tilde{D}^{-1/2}\tilde{R}\tilde{D}^{-1/2}, \qquad \tilde{D} = \mathrm{diag}(\tilde{R}).$$

This is a rigorous prior on correlation matrices and is close to the inverse-Wishart calculations we have already used. For large ν_0, it should give approximately Gaussian local fluctuations around R_0. However, the induced distribution of R is not inverse-Wishart, the column marginals are not exactly Gaussian, and $E(R)$ is not necessarily exactly R_0. Thus $R_0 = \bar{R}$ should initially be interpreted as a centering matrix, not as an exact prior mean unless calibrated.

LKJ-type prior centered at $\bar{R}$. The LKJ prior is a native prior on correlation matrices, with a concentration parameter, but the standard form is centered at the identity matrix. One could consider a transformed or tilted LKJ-type prior centered near $\bar{R}$. This has clean support but may not lead to simple Schur-complement column updates.

Partial-correlation or vine parameterization. Correlation matrices can be parameterized by partial correlations. We could put shrinkage priors on those partial correlations, centered at the corresponding partial correlations of $\bar{R}$. This gives exact support and flexible local uncertainty. The drawback is that the induced distribution on ordinary LD columns $R_{-j,j}$ may be algebraically awkward.

Matrix-log or tangent-space prior. Another possibility is to map correlation matrices into an unconstrained tangent-like space, put a Gaussian prior around a transform of $\bar{R}$, and map back to $\mathcal{C}_J$. This is appealing for local perturbation theory, but the Jacobian and the induced column conditionals may be inconvenient for inference.

Projected Gaussian perturbation. We could perturb $\bar{R}$ by a Gaussian symmetric matrix and project the result back to $\mathcal{C}_J$. This may be computationally convenient, but the projection creates a density and dependence structure that is hard to interpret as a clean hierarchical model.

At this stage, the most promising strategy is:

1. Use a proper distribution on $\mathcal{C}_J$, such as standardized inverse-Wishart, to make the hierarchical model rigorous.
2. Use the exact ideal-PIP identity, which is prior-agnostic as long as $\mathrm{diag}(R) = 1$.
3. Use Gaussian Schur-complement column updates as local variational approximations for computation, rather than claiming the exact prior has Gaussian column marginals.

6 standardized IW

We consider

$$\tilde{R} \sim \mathcal{IW}(\nu_0 \bar{R}, \nu_0 + J + 1), \qquad R = \tilde{D}^{-1/2} \tilde{R} \tilde{D}^{-1/2}, \qquad \tilde{D} = \mathrm{diag}(\tilde{R}).$$

Let $p_{\mathrm{SIW}}(R \mid \bar{R}, \nu_0)$ denote the induced distribution on correlation matrices. Still consider the SER model:

$$\begin{aligned} \gamma &\sim \mathrm{Unif}\{1, \ldots, J\}, \qquad \beta \sim \mathcal{N}(0, \sigma_0^2), \\ x \mid R, \beta, \gamma = j &\sim \mathcal{N}(\beta R_{\cdot j}, R/N). \end{aligned} \tag{2}$$

Exact posterior factorization

Let

$$p_0(x \mid R) = \mathcal{N}(x \mid 0, R/N).$$

Using the likelihood-ratio identity from the ideal-PIP derivation,

$$p(x \mid R, \beta, \gamma = j) = p_0(x \mid R) \exp\left\{ N\beta x_j - \frac{N}{2}\beta^2 \right\}.$$

Therefore

$$p(R, \beta \mid x, \gamma = j) \propto p_{\mathrm{SIW}}(R \mid \bar{R}, \nu_0) p_0(x \mid R)\, p(\beta) \exp\left\{ N\beta x_j - \frac{N}{2}\beta^2 \right\}.$$

Thus R and β factorize in the posterior conditional on $\gamma = j$:

$$p(R, \beta \mid x, \gamma = j) = p(R \mid x)\, p(\beta \mid x_j),$$

where

$$p(R \mid x) \propto p_{\mathrm{SIW}}(R \mid \bar{R}, \nu_0) \mathcal{N}(x \mid 0, R/N)$$

is common for all j, and

$$\beta \mid x_j, \gamma = j \sim \mathcal{N}(m_j, v), \qquad v = (N + \sigma_0^{-2})^{-1}, \qquad m_j = vNx_j = \frac{N\sigma_0^2}{1 + N\sigma_0^2} x_j.$$

PIP

Integrating out β gives the usual single-effect approximate Bayes factor

$$\mathrm{ABF}_j = (1 + N\sigma_0^2)^{-1/2} \exp\left\{ \frac{N}{2} \frac{N\sigma_0^2}{1 + N\sigma_0^2} x_j^2 \right\}.$$

The R-dependent factor

$$\int p_{\mathrm{SIW}}(R \mid \bar{R}, \nu_0) \mathcal{N}(x \mid 0, R/N)\, dR$$

is common across j. Therefore

$$\Pr(\gamma = j \mid x) = \frac{\mathrm{ABF}_j}{\sum_{k=1}^{J} \mathrm{ABF}_k}.$$

Thus the standardized-IW prior preserves the ideal SER PIP exactly.

Posterior of R

The posterior for R is rigorous but not conjugate in the standardized-IW parameterization. In terms of $\tilde{R}$,

$$p(\tilde{R} \mid x) \propto \mathcal{IW}(\tilde{R}; \nu_0 \bar{R}, \nu_0 + J + 1) |R(\tilde{R})|^{-1/2} \exp\left\{ -\frac{N}{2} x^\top R(\tilde{R})^{-1} x \right\},$$

where

$$R(\tilde{R}) = \tilde{D}^{-1/2} \tilde{R} \tilde{D}^{-1/2}.$$

Equivalently,

$$|R(\tilde{R})| = \frac{|\tilde{R}|}{\prod_{k=1}^{J} \tilde{R}_{kk}}, \qquad R(\tilde{R})^{-1} = \tilde{D}^{1/2} \tilde{R}^{-1} \tilde{D}^{1/2}.$$

The dependence of $\tilde{D}$ on $\tilde{R}$ breaks inverse-Wishart conjugacy. Exact inference for R would therefore require numerical integration, MCMC, Laplace approximation, or another approximation.

Column inference

For relaxed-column use, define

$$z_j = R_{-j,j}, \qquad S_j = R_{-j,-j} - z_j z_j^\top.$$

Under the common posterior $p(R \mid x)$, the marginal posterior for the j-th column can be written using the decomposition above:

$$p(z_j, S_j \mid x) \propto p_{\mathrm{SIW},j}(z_j, S_j \mid \bar{R}, \nu_0) \mathcal{N}\left(x_{-j} \mid x_j z_j, \frac{1}{N} S_j \right),$$

where $p_{\mathrm{SIW},j}$ is the marginal prior induced by standardized IW on the block variables (z_j, S_j). The factor $\mathcal{N}(x_j \mid 0, 1/N)$ is constant with respect to (z_j, S_j) and has been dropped.

This expression is exact, but $p_{\mathrm{SIW},j}(z_j, S_j \mid \bar{R}, \nu_0)$ does not have the same simple normal–inverse-Wishart form as the unstandardized inverse-Wishart prior. A practical local approximation is to use the familiar Schur-complement working prior

$$z_j \mid S_j \approx \mathcal{N}(\mu_j, S_j/\nu_{\mathrm{eff}}), \qquad \mu_j = \bar{R}_{-j,j},$$

with ν_{eff} calibrated to the standardized-IW concentration. Then

$$z_j \mid S_j, x \approx \mathcal{N}\left(\frac{\nu_{\mathrm{eff}}\mu_j + N x_j x_{-j}}{\nu_{\mathrm{eff}} + N x_j^2}, \frac{S_j}{\nu_{\mathrm{eff}} + N x_j^2}\right).$$

Equivalently, the approximate posterior mean column is

$$\mathbb{E}(z_j \mid x) \approx \mu_j + \frac{N x_j}{\nu_{\mathrm{eff}} + N x_j^2}(x_{-j} - x_j \mu_j).$$

The exact model supplies the ideal PIP and a proper correlation-matrix prior; the Gaussian column update is then a local variational approximation used for computing relaxed effect shapes.